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Taishi Nakamura

Education

- 04/2026– **PhD in Computer Science**, *Institute of Science Tokyo*, Tokyo, Japan
Present Supervisor: Rio Yokota
- 04/2024– **MS in Computer Science**, *Institute of Science Tokyo*, Tokyo, Japan
03/2026 Supervisor: Rio Yokota
- 04/2021– **BS in Computer Science**, *Tokyo Institute of Technology*, Tokyo, Japan
03/2024 Completed degree one year early in recognition of academic excellence

Employment

- 05/2024– **Research Assistant**, *National Institute of Informatics Research and Development*
present *Center for Large Language Models*
Working with Prof. Jun Suzuki and Prof. Rio Yokota
- 02/2024– **Research Intern**, *Sakana AI*, Tokyo, Japan
present Working with Dr. Takuya Akiba
- 08/2023– **Student Trainee**, *National Institute of Advanced Industrial Science and Technology*
present Working with Dr. Hiroya Takamura,
Prof. Naoaki Okazaki, Prof. Rio Yokota
- 04/2023– **Research Assistant**, *Institute of Science Tokyo*, Tokyo, Japan
present Working with Prof. Rio Yokota
- 10/2023– **Research Intern**, *LLM-jp Project*
04/2024 Working with Prof. Jun Suzuki
- 06/2023– **MITOU Target Program Fellow**, *Information-technology Promotion Agency*
02/2024 Mentor: Dr. Yuuki Tokunaga

Research Interests

I am interested in efficient training of large language models, including scaling and training Mixture-of-Experts (MoE) models, continual pre-training, and efficient post-training with reinforcement learning algorithms.

Selected Awards and Grants

- 2025 **Language Resources Award**, Annual Meeting of the Japanese Association for NLP
- 2024 **Outstanding Paper Award**, Annual Meeting of the Japanese Association for NLP

Projects

LLM-jp – Core Contributor. LLM-jp is a fully open, scratch-trained series of large language models primarily focused on Japanese. The project emphasizes transparency in data, training, and model design. Contributed to the design and large-scale training of the LLM-jp models. For LLM-jp-3, applied a novel sparse Mixture-of-Experts training strategy (Drop-Upcycling; ICLR 2025) to scale the model to $8 \times 13\text{B}$ parameters on 256 GPUs, achieving superior performance compared to a 172B dense model trained on the same data. At the time, it was among the most competitive open Japanese LLMs. For LLM-jp-4, designed the model architecture and training strategy based on findings from our ICLR 2026 work on optimal sparsity in MoE language models.

Swallow LLM – Swallow LLM is a series of Japanese-specialized large language models developed through continual pre-training. Co-led the early development of Swallow LLM, establishing effective continual pre-training methods for Japanese LLM adaptation (COLM 2024) and designing synthetic instruction-tuning data generation pipelines for early-stage alignment. Led large-scale continual pre-training and post-training of Gemma-2 Swallow on TPU v6e (256 chips), achieving state-of-the-art Japanese performance in the 2B and 9B model classes. The 27B model demonstrated performance comparable to 70B-scale models. More recently, led reinforcement learning-based post-training for Qwen3-Swallow and GPT-OSS Swallow to further improve reasoning capabilities.

Aurora-M (COLING 2025) – Led Aurora-M, an open-source multilingual and code-focused continual pre-training project. Coordinated cross-institutional collaboration and executed large-scale distributed training on the LUMI supercomputer (32 nodes, AMD GPUs). Contributed to open-science initiatives for multilingual and code LLM development.

Publications

Peer-Reviewed Conference Publications

Kazuki Fujii, Yukito Tajima, Sakae Mizuki, Masaki Kawamura, Hinari Shimada, Taihei Shiotani, Koshiro Saito, Masanari Oi, **Taishi Nakamura**, Takumi Okamoto, Shigeki Ishida, Kakeru Hattori, Youmi Ma, Hiroya Takamura, Rio Yokota, Jun Sakuma, and Naoaki Okazaki. Rewriting Pre-Training Data Boosts LLM Performance in Math and Code. In *International Conference on Learning Representations (ICLR)*, 2026.

Huu Nguyen*, Victor May*, Harsh Raj*, Marianna Nezhurina, Yishan Wang, Yanqi Luo, Vu Minh Chien, **Taishi Nakamura**, Ken Tsui, Van Khue Nguyen, David Salinas, Aleksandra Krasnodebska, Christoph Schuhmann, Mats Leon Richter, Xuan-Son Vu, and Jenia Jitsev. MixtureVitae: Open Web-Scale Pretraining Dataset With High Quality Instruction and Reasoning Data Built from Permissive-First Text Sources. *Transactions on Machine Learning Research*, 2026. **Featured Certificate**

tion, J2C Certification.

Taishi Nakamura, Satoki Ishikawa, Masaki Kawamura, Takumi Okamoto, Daisuke Nohara, Jun Suzuki, and Rio Yokota. Optimal Sparsity of Mixture-of-Experts Language Models for Reasoning Tasks. In *International Conference on Learning Representations (ICLR)*, 2026. [Oral](#).

So Kuroki, **Taishi Nakamura**, Takuya Akiba, and Yujin Tang. Agent Skill Acquisition for Large Language Models via CycleQD. In *International Conference on Learning Representations (ICLR)*, 2025.

Youmi Ma, Sakae Mizuki, Kazuki Fujii, **Taishi Nakamura**, Masanari Ohi, Hinari Shimada, Taihei Shiotani, Koshiro Saito, Koki Maeda, Kakeru Hattori, et al. Building Instruction-Tuning Datasets from Human-Written Instructions with Open-Weight Large Language Models. In *Conference on Language Modeling (COLM)*, 2025.

Taishi Nakamura^{*}, **Mayank Mishra**^{*}, **Simone Tedeschi**^{*}, Yekun Chai, Jason T. Stillerman, Felix Friedrich, Prateek Yadav, Tanmay Laud, Vu Minh Chien, Terry Yue Zhuo, et al. Aurora-M: Open Source Continual Pre-training for Multilingual Language and Code. In *International Conference on Computational Linguistics (COLING): Industry Track*, 2025.

Taishi Nakamura, Takuya Akiba, Kazuki Fujii, Yusuke Oda, Rio Yokota, and Jun Suzuki. Drop-Upcycling: Training Sparse Mixture of Experts with Partial Re-initialization. In *International Conference on Learning Representations (ICLR)*, 2025.

Yuichi Inoue^{*}, Kou Misaki^{*}, Yuki Imajuku, So Kuroki, **Taishi Nakamura**, and Takuya Akiba. Wider or Deeper? Scaling LLM Inference-Time Compute with Adaptive Branching Tree Search. In *International Conference on Neural Information Processing Systems (NeurIPS)*, 2025. [Spotlight](#).

Kazuki Fujii^{*}, **Taishi Nakamura**^{*}, Mengsay Loem, Hiroki Iida, Masanari Ohi, Kakeru Hattori, Hirai Shota, Sakae Mizuki, Rio Yokota, and Naoaki Okazaki. Continual Pre-Training for Cross-Lingual LLM Adaptation: Enhancing Japanese Language Capabilities. In *Conference on Language Modeling (COLM)*, 2024. [Outstanding Paper & Language Resource Award at Annual Meeting of the Japanese Association for NLP](#).

Naoaki Okazaki, Kakeru Hattori, Hirai Shota, Hiroki Iida, Masanari Ohi, Kazuki Fujii, **Taishi Nakamura**, Mengsay Loem, Rio Yokota, and Sakae Mizuki. Building a Large Japanese Web Corpus for Large Language Models. In *Conference on Language Modeling (COLM)*, 2024. [Outstanding Paper Award at Annual Meeting of the Japanese Association for NLP](#).

Workshop Publications

Marianna Nezhurina^{*}, Jörg K.H. Franke, **Taishi Nakamura**, Timur Carstensen, Niccolò Ajroldi, Ville Komulainen, David Salinas, and Jenia Jitsev^{*}. open-sci-ref-0.01: open and reproducible reference baselines for language model and dataset comparison. In *AAAI Workshop on Reproducible AI*, 2025.

Koshiro Saito, Sakae Mizuki, Masanari Ohi, **Taishi Nakamura**, Taihei Shiotani, Koki Maeda, Youmi Ma, Kakeru Hattori, Kazuki Fujii, Takumi Okamoto, et al. Why We Build Local Large Language Models: An Observational Analysis from 35 Japanese and Multilingual LLMs. In *Conference on Language Modeling (COLM) Multilingual and Equitable Language Technologies Workshop*, 2025. **Outstanding Paper Award, Natural Language Processing Research Meeting of the Information Processing Society of Japan.**

Kazuki Fujii, **Taishi Nakamura**, and Rio Yokota. Ilm-recipes: A Framework for Seamless Integration and Efficient Continual Pre-Training of Large Language Models. In *International Conference for High Performance Computing, Networking, Storage, and Analysis (SC) Trillion Parameter Consortium Workshop*, 2024.

Preprints

Daisuke Nohara, **Taishi Nakamura**, and Rio Yokota. On the Optimal Reasoning Length for RL-Trained Language Models. arXiv:2602.09591, 2026.

Kazuki Fujii, **Taishi Nakamura**, and Rio Yokota. Balancing Speed and Stability: The Trade-offs of FP8 vs. BF16 Training in LLMs. arXiv:2411.08719, 2024.

LLM-jp. LLM-jp: A Cross-organizational Project for the Research and Development of Fully Open Japanese LLMs. arXiv:2407.03963, 2024.

Service

Reviewer

2025–present **NeurIPS 2026, ICLR 2026**

2025–present **Workshops, ICML 2026 AdaptFM, ICML 2025 AI4Math, NeurIPS 2025 ER**